

Nikhil Sobanbabu

Curriculum Vitae

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Education

Carnegie Mellon University

Pittsburgh, PA

MASTER OF SCIENCE IN ELECTRICAL AND COMPUTER ENGINEERING **GPA: 4.0/4.0**

Aug 2024 – May 2026

Coursework: Robot Learning, Optimal Control, Deep Learning for Computer Vision, Optimization, Planning and Decision Making.

Indian Institute of Technology Madras

Chennai, India

BACHELOR OF TECHNOLOGY IN ELECTRICAL ENGINEERING **GPA: 9.39/10.0**

Aug 2020 – May 2024

Coursework: Reinforcement Learning, Motion Planning, Linear Dynamical Systems, Nonlinear System Analysis, Field Robotics

Research Interests

My research goal is to develop physics-aware learning and control frameworks that enable robots to acquire scalable, contact-rich skills and execute them robustly in the real world. I am particularly interested in **Learning-based control** (Agile Locomotion, Loco-Manipulation and Human Motion tracking), **Real2Sim2Real** (System Identification, Dynamics-Aware sim-to-real adaptation, Active Exploration, reality-gap aware benchmarks) and **Multi-Robot coordination** spanning algorithms and training pipelines.

Honors and Awards

- 2025 **CoRL ORAL**, Top 5.7%
- 2024 **KC Mahindra Graduate Scholarship**, Selected 90/2354
- 2023 **IUSSTF-Viterbi Summer Research Internship Program**, Selected 15/400
- 2020 **All India Rank 286 in JEE Advanced**, 0.2% among 1.2 Million Students
- 2020 **KVPY Fellow SX**, All India Rank of 243.
- 2018 **NTSE Scholar**, State Topper, Stage 1

Publications

PREPRINTS AND UNDER REVIEW

HDMI: Learning Interactive Humanoid Whole-Body Control from Human Videos

Haoyang Weng, Yitang Li, **Nikhil Sobanbabu**, Zihan Wang, Zhengyi Luo, Tairan He, Deva Ramanan, Guanya Shi
arXiv preprint, 2025, [Paper]

OGMP: Oracle Guided Multi-mode Policies for Agile and Versatile Robot Control

Lokesh Krishna, **Nikhil Sobanbabu**, Quan Nguyen
arXiv preprint, [Paper]

CONFERENCE PROCEEDINGS

* Equal contribution.

FADA: Few-Shot Domain Adaptation via Dynamics Alignment for Humanoid Control

Angchen Xie*, **Nikhil Sobanbabu***, Ishayu Shikhare, Alan Wang, Max Simchowitz, Guanya Shi
Conference on Robot Learning (CoRL 2026, accepted), [Paper]

RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning

Pablo Ortega-Kral*, Eliot Xing*, Arthur Buckner, Vernon Luk, Junseo Kim, Owen Kwon, Angchen Xie, **Nikhil Sobanbabu**, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussell, Guanya Shi, Jonathan Francis, Jean Oh
Robotics: Science and Systems (RSS 2026), [Paper]

Sampling-Based System Identification with Active Exploration for Legged Robot Sim2Real Learning

Nikhil Sobanbabu*, Guanqi He*, Tairan He, Yuxiang Yang, Guanya Shi (**Perfect Score-All Strong Accepts**)
Conference on Robot Learning (CoRL 2025 (Oral)), [Paper]

ASAP: Aligning Simulation and Real-World Physics for Learning Agile Humanoid Whole-Body Skills

Tairan He, Jiawei Gao, Wenli Xiao, Yuanhang Zhang, Zi Wang, Jiashun Wang, Zhengyi Luo, Guanqi He, **Nikhil Sobanbabu**, Chaoyi Pan, Zeji Yi, Guannan Qu, Kris Kitani, Jessica Hodgins, Linxi “Jim” Fan, Yuke Zhu, Changliu Liu, Guanya Shi
Robotics: Science and Systems (RSS 2025), [Paper]

Preferred Oracle Guided Multi-Mode Policies for Dynamic Bipedal Loco-Manipulation

Prashanth Ravichandar*, Lokesh Krishna*, **Nikhil Sobanbabu**, Quan Nguyen
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025), [Paper]

Towards Unstructured MAPF: Multi-Quadruped MAPF Demo

Rishi Veerapaneni*, **Nikhil Sobanbabu***, Guanya Shi, Jiaoyang Li, Maxim Likhachev
International Conference on Automated Planning and Scheduling (ICAPS 2025, Demo Track), [Paper]

Research Experience

Carnegie Mellon University, LeCAR Lab

GRADUATE RESEARCH ASSISTANT, ADVISED BY **PROF. GUANYA SHI**

Pittsburgh, PA

May 2024 – May 2026

- Co-developed **FADA**, a planner-inverse dynamics framework for few-shot humanoid adaptation that fine-tunes the inverse dynamics model using approximately two minutes of target-domain rollouts. Co-first author; CoRL 2026 (accepted).
- Developed a parallelizable **sampling-based system identification (SPI-Active)** framework that identifies inertial and actuator parameters of legged robots by reducing open-loop trajectory prediction error enabling precise locomotion.
- Designed an **active exploration policy** maximizing Fisher Information to enable informative data collection for parameter estimation, and deployed agile RL locomotion policies, improving transfer performance by **42–63%** over baselines.

Carnegie Mellon University, Search-Based Planning Lab (SBPL)

GRADUATE RESEARCH ASSISTANT, ADVISED BY **PROF. MAXIM LIKHACHEV**

Pittsburgh, PA

Jan 2025 – May 2026

- Worked on **multi-agent motion planning and coordination** for teams of quadruped robots using Conflict-Based-Search(CBS) and Temporal Graph(TPG)-based planning.
- Developed a waypoint refinement and synchronization pipeline together with custom-trained locomotion policies to enable coordinated navigation of heterogeneous quadruped robots.
- Developed a modular sim-to-real control stack for Unitree quadrupeds, enabling scalable multi-agent deployment and seamless hot-swapping between high-level and low-level controllers.

University of Southern California, Dynamic Robotics and Control Lab (DRCL)

RESEARCH INTERN, ADVISED BY **PROF. QUAN NGUYEN**

Los Angeles, CA

May 2023 – May 2026

- Developed a **closed-loop state reference generator(oracle)** using preview control and a centroidal model that generates dynamically feasible state trajectories guiding an RL policy for the biped Hector toward **parkour-style motions** in MuJoCo.
- Extended the framework with a **hybrid automaton-based oracle** that supports multi-mode, whole-body loco-manipulation behaviors such as soccer dribbling and box manipulation across morphologies(HECTOR v1, Berkeley Humanoid, Unitree G1).
- Worked on a **data-augmentation pipeline** to learn oracles directly from large-scale retargeted human motion data and automate oracle design.

Indian Institute of Technology Madras, Control Engineering Laboratory

UNDERGRADUATE RESEARCH ASSISTANT, ADVISED BY **PROF. BHARATH BHIKKAJI**

Chennai, India

Aug 2022 – May 2024

- Studied single-pursuer, multi-evader **pursuit-evasion-target** games for my undergraduate thesis, deriving optimal reach-avoid strategies using **HJI-PDE** equations for omnidirectional agents and implemented them on LEGO EV3 platform for three agents.
- Formulated a Multi-agent RL framework using **MAPPO** on a custom **petting-zoo** environment and compared the convergence with nash-equilibrium of analytical solution on a 1v1 game.

Industry Experience

Amazon FAR

MTS INTERN

May 2026 – Present

- Developing cross-trial behavior adaptation for humanoid loco-manipulation tasks.
- Integrating Fauna Sprout with Holosoma.

DiFACTO Robotics and Automation

ROBOTICS INTERN

Bengaluru, India

Jun 2024 – Aug 2024

- Developed a natural-feature navigation and recovery framework for in-house AMRs using ROS NavStack
- Optimized **Fanuc R-1000iA/80H** palletization trajectories, reducing cycle time by **23.6%**.

Skills

Programming	C/C++, Python, Julia, Bash, Git
Frameworks & Tools	PyTorch, TensorFlow, CasADi, NumPy, Pandas, scikit-learn, Docker, GitHub
Simulation & Robotics	Isaac Gym/Sim, MuJoCo, ROS1/2, Gazebo, MoveIt, MATLAB/Simulink
Domains	Reinforcement Learning, Optimal & Adaptive Control, Sim2Real Transfer
Hardware & Electronics	Embedded Systems (STM32, Arduino, Jetson), Altium PCB Design, Sensors & Actuators Integration

Academic Services

Reviewer	ICRA, IROS, RAL, CoRL 2025, ICRA 2026
Teaching Assistant	CMU 18-460 <i>Feedback Control Systems</i> , CMU 18-202 <i>Mathematical Foundations for Electrical Engineers</i>
Teaching Assistant	EE3004 Control Engineering

Media Coverage

IEEE Spectrum – Video Friday(Evan Ackerman)

OGMP: Oracle Guided Multimodal Policies for Agile and Versatile Robot Control [\[Link\]](#)

Projects and Competitions

Adaptive Force Control for Learned Centroidal Controller

COURSE PROJECT (OPTIMAL CONTROL AND RL) [GITHUB]

Carnegie Mellon University

Spring 2025

- Designed a control framework integrating **L1-Adaptive force control** with a **Hierarchical RL and QP-based** low-level controller.
- Achieved **2.5–3x** longer traversal distances in a continuous jumping task via adaptive sim-to-sim transfer enabling real-time payload adaptation for quadruped robots.

Terrain Traversability Analysis with Image Dehazing

COURSE PROJECT (DEEP LEARNING FOR CV) [REPORT]

Carnegie Mellon University

Fall 2024

- Integrated a Generative Adversarial Network (GAN)-based dehazing model with a terrain segmentation (GANav) framework improving mIoU by **61.29%**.
- Analysed and compared Physics and Learning based dehazing techniques as an input to the GANav framework.

Motion Imitation with DIAL-MPC

COURSE PROJECT (ROBOT LEARNING) [REPORT]

Carnegie Mellon University

Fall 2024

- Generated reference trajectories for Unitree Go2 using motion retargeting from motion capture data of dogs.
- Implemented a Full-order Sampling based MPC framework enhancing trajectory tracking performance by 22.73%.

Multi-Agent Trajectory Tracking for Crazyflie Quadrotors

UNDERGRADUATE RESEARCH PROJECT [REPORT]

IIT Madras

2023

- Worked on **autonomous** control of multiple **crazyflie quadrotors** using custom-built **ROS** packages with modified **Mellinger and PID** controllers under the guidance of Dr. Bharath Bikkaji.
- Created **ROS** meta packages for **safe landing** of the **quadrotors** in the event of loss of communication.

Team Anveshak

TEAM LEAD

IIT Madras

2021 – 2023

- Designed custom **PCBs** utilizing **MPC2515 CAN** controller to facilitate precise control of the actuators in the manipulator of the rover via **Nvidia's Jetson Xavier**.
- Engineered an energy-efficient power distribution and monitoring framework, reducing rover energy consumption to **86%**. Enabled remote monitoring of electrical parameters from a distant base station, enhancing rover performance.
- Led the team of 40+ students and secured a **global 6th** place at the **Anatolian Rover Challenge in Turkey, 2022**.

Swing-up and Stabilization of an Inverted Pendulum

COURSE PROJECT (NONLINEAR SYSTEM ANALYSIS)

IIT Madras

Spring 2023

- Developed a controller for the **swing-up and stabilization** of a linear inverted pendulum.
- Implemented swing-up control using **energy-based Lyapunov functions** and stabilized via **pole placement** after swing-up.
- Analyzed robustness and performance of a **Sliding Mode Controller** for stability improvement.

Sampling-Based Motion Planning for a Mobile Manipulator

COURSE PROJECT (MOTION PLANNING)

IIT Madras

Fall 2022

- Implemented sampling-based planners (**RRT*** and **Bi-Directional RRT**) for the KUKA mobile manipulator.
- Integrated motion planning with **A*** and **Traveling Salesman** algorithms for optimal pick-and-place tasks in Gazebo.
- Validated task completion efficiency through simulations with dynamic obstacle configurations.

KrishiBot – Autonomous Agriculture Robot

COMPETITION PROJECT (EYRC 2022–23)

IIT Madras

Oct 2022 – Mar 2023

- Implemented a **wall-following navigation algorithm** using laser scan feedback and a PID controller for warehouse traversal.
- Integrated **color and centroid detection** to identify bell peppers using Intel RealSense depth camera.
- Executed a complete **pick-and-place pipeline** in ROS MoveIt for crop harvesting automation.